

Research Report

AUTHOR

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library(dplyr)
library(DiagrammeR)
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Introduction

Researcher degrees of freedom describe the flexibility that researchers have in decision making during data collection and analysis. This includes decisions such as how many participants to collect, which statistical model to use and which covariates to add. These choices need to be made at some point in the research process, but *when/at what point in the process* these decisions are made can impact the quality of research. It commonly occurs that researchers make decisions during data collection or analysis, instead of at the beginning of the research process. This can cause researchers to, knowingly or unknowingly, steer their research towards desired outcomes. In fact, researcher degrees of freedom are thought to be an important part of why it is hard to reproduce many findings in psychology ([Simmons, Nelson, and Simonsohn 2011b](#)).

When researchers engage in this type of decision making it is often referred to as 'p-hacking' ([Stefan and Schönbrodt 2023](#)). Common p-hacking strategies are choices such as selective reporting of the dependent and independent variables, optional stopping and outlier exclusion. These decisions can all impact the false positive rates in research. Simmons, Nelson, and Simonsohn ([2011a](#)) similarly showed how decisions such as including interaction effects or adding additional participants can lead to inflated false positive rates. Decisions made during data collection and analysis can thus impact research results in unintended ways, often making it harder to reproduce and corroborate findings.

Preregistrations were proposed as a solution to this problem. Preregistrations are documents in which researchers provide their data collection and analysis plan. These plans are then published before data collection and ideally, they are as detailed as possible. By securing these decisions beforehand, flexibility in research choices during collection and analysis is limited with the aim to improve reproducibility and prevent false-positive inflation.

Over the years there has been an increase in the use of preregistrations ([Lindsay and Stephen 2018](#); [P. Simmons, D. Nelson, and Simonsohn 2021](#)). However, it remains unclear whether the use of preregistration actually limits flexibility and improves reproducibility. Akker et al. ([2024](#)) found that preregistered papers do not differ in the amount of significant results published and they contain the same amount of statistical errors.

The lack of evidence for the effectivity of preregistrations might be due to deviations from preregistrations. Researchers who preregister their research also commonly deviate from them. An estimated 82-93% of papers contain deviations of which only 55-63% are actually reported ([Willroth and Atherton 2024](#); [Claesen et al. 2021a](#)). This means that researchers register their plan beforehand but end up changing their choices during the execution. Deviations occur in all aspects of the research process,

such as the variable operationalization, data collection procedure, statistical analysis and outlier identification ([Van Den Akker et al. 2024a](#)). The consistency between preregistrations plans and the methodology in published papers can be as low as 28% in some of these categories. According to ([vandenakker?](#)), the most common deviations occur in data collection procedures, determining exclusion criteria and choosing the statistical model.

However, deviations are not inherently a bad thing. Preregistrations were always meant to be a framework, not hard law ([Hardwicke and Wagenmakers 2023](#)). Deviating can be done opportunistically as a form of p-hacking but can also be due to necessity or to improve the robustness of the research ([Lakens, n.d.](#)). However, little is known about why researchers choose to deviate.

What has not been researched either is the effect of deviations from preregistrations. Lakens ([n.d.](#)) argues that, depending on the deviation and the motivation, deviations can have either a positive or negative effect on the severity of a test. This means that changing the preregistered research plan could lead to better detection of existing effects and better rejection of false effects, but also to less robustly tested research. No actual studies have shown these theorized effects however. Which raises the question: "What are the actual effects of deviations from preregistrations?"

Furthermore, do these deviations matter in the first place? Willroth and Atherton ([2024](#)) showed that researchers evaluate publication with deviations based on many factors. Transparency of the authors was deemed most important but the number of deviations less so. Some types of deviations also seemed to be more or less concerning than others.

With this simulation study, I intend to close these gaps in the literature and explore the effects of typical preregistration deviations on research outcomes and the quality of published results. Specifically, I aim to answer the question: 'How do deviations from preregistration affect type I and type II error rates in research?' This thesis will investigate this question in the context of behavioral psychology. The aim of this study is descriptive and therefore no specific hypotheses will be formalized.

Based on how commonly they occur, their potential impact, and their justifiability, four important deviations were identified: sample size, outlier identification, hypotheses and the statistical model. Through a simulation study this thesis shows how deviations in each of these categories impact type I and type II error rates.

Methods/ADEMP

Conditions/Deviations

I identified four important types of deviations to investigate based on (1) how often they occur, (2) their potential impact and (3) their justifiability. The deviations categories are: sample size, outlier identification, hypotheses and statistical model

Sample size

Deviations in the sampling size are some of the most common deviations, with consistency between preregistrations and published papers being 28% for the exact sampling size ([Akker et al. 2024](#)). Changing the sample size can decrease power in research as well as inflate the false-positive rates ([Lakens, n.d.](#); [Simmons, Nelson, and Simonsohn 2011c](#)). Small deviations in sample size are commonly due to oversights during the collection process. Researchers also admit to stopping collection earlier or

continuing collection longer after finding disappointing results (John, Loewenstein, and Prelec 2012), however, whether this is also used as a reason to deviate from preregistrations is unknown. Based on this, the following deviations in sample size will be simulated.

Table 1: Sample Size Deviations

Type of deviation	Deviations			
Increase	1%	10%	25%	50%
Decrease	1%	10%	25%	50%

Outlier identification

Literature shows that over half of published papers do not adhere to their own specified outlier criteria (Van Den Akker et al. 2024b; Claesen et al. 2021b). Reasons for these deviations often include vague criteria in the preregistration or failing to specify outlier criteria at all in preregistrations. When criteria are specified, they often leave a lot of room for interpretation and ad hoc decisions on exclusion (Heirene et al. 2024). However, Bakker and Wicherts (2014) showed that there is no significant difference in the amount of significant findings between papers with outlier exclusion and those without. This suggests that adding outlier exclusion criteria when they were not specified in the preregistration might not be a problem. Researchers also deem outlier deviations to be relatively acceptable (Willroth and Atherton 2024).

However, Lakens (n.d.) argues that adding additional exclusion criteria can undermine the severity of a test. This idea is corroborated by Simmons, Nelson, and Simonsohn (2011c), who showed that the Type I error increases linearly with the number of outlier methods used. Furthermore, 38% of researchers admit to excluding outliers only after examining their impact on the data (John, Loewenstein, and Prelec 2012). If this is also the reason why many people deviate from their preregistered outlier criteria, then these deviations could be impactful. Consequently, whilst deviations in outlier identification are common and relatively accepted, they might have quite an effect. Based on this, the starting point will be to not exclude any datapoints and to simulate the following deviations.

Table 2: Outlier Identification Deviations

Method	Outlier criterium
Z-score	> 3 sd
Cook’s Distance	> 1

Hypotheses

Akker et al. (2023) found that 60% of published papers contain a difference in hypotheses when compared to their corresponding preregistration. However, there is no evidence that papers with or without deviations in their hypotheses report more significant results. The deviations that occur are varied and are often due either to improper specification in the preregistration (Heirene et al. 2024). Another reason that researchers might choose to alter their hypotheses during their research is to be able to select hypotheses with desired results (John, Loewenstein, and Prelec 2012). Even though they occur frequently, changes in the hypotheses are ranked as the least justifiable deviations by researchers themselves (Willroth and Atherton 2024). The following deviations will be simulated.

Table 3: Hypothesis Deviations

Deviation
changed directionality
subgrouping*

Statistical model

Amongst all of the aforementioned categories, the least is known about why people deviate from their preregistered model. Different papers/research investigates deviations on different aspects of statistical models.

Table 4: Statistical Model Deviations

Variable	Deviation
Dependent variable	Scaling the variable
Covariate	Add continuous covariate
	Add dichotomous covariate

Aims

The aim is to examine the increase/decrease in type I and type II errors.

Data-generating mechanism

Data is generated under two scenarios; the null-hypothesis is true and the alternative hypothesis is true. Both scenarios generate data by producing parametric draws from a linear regression model. As such, the null-hypothesis and alternative hypothesis will be as follows:

$H0 : \beta_{xy} = 0$

$Ha : \beta_{xy} \neq 0$

OR OR OR we do a verbal hypothesis *H1: X has an effect on Y*. The dependent variable will be generated in accordance to the standard regression formula:

$Y = \beta_0 + \beta_1 X + \epsilon$

Data will be generated under the conditions in **?@tbl-DGM**.

Table 5: Data generating conditions

parameter/scenario	$H0 = true$	$Ha = true$
Independent variable	$X \sim \mathcal{N}(\mu, \sigma^2)$	$X \sim \mathcal{N}(\mu, \sigma^2)$
Random error	$\epsilon \sim \mathcal{N}(\mu, \sigma^2)$	$\epsilon \sim \mathcal{N}(\mu, \sigma^2)$
Regression coefficient	$\beta_1 = 0$	$\beta_1 = 0.30$

parameter/scenario	$H0 = true$	$Ha = true$
Intercept	$\beta_0 = TBD$	$\beta_0 = TBD$
Continuous demographic variable	$Z \sim \mathcal{N}(\mu, \sigma^2)$	$Z \sim \mathcal{N}(\mu, \sigma^2)$
Dichotomous demographic variable	$D \sim \text{Bernoulli}(0.5)$	$D \sim \text{Bernoulli}(0.5)$
Sample size	150	150

Estimands

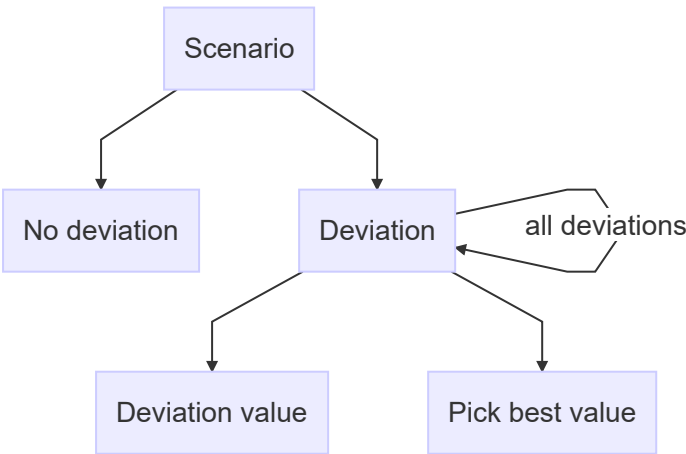
nominal type I, type II Accepted type I error is based on an alpha level of 0.05. The nominal type II error is 0.20, resulting in a power of 80%.

Methods

Ranges from no deviation - all deviations.

Deviation scenarios

```
DiagrammeR("
graph TD;
  A[Scenario] --> F[No deviation];
  A --> G[Deviation];
  G --> |all deviations| G;
  G --> J[Deviation value];
  G --> K[Pick best value];
")
```



Performance measures

Actual Type I and Type II error

bias in regression coefficient?

Results

- include what I expect results to look like

Discussion

- limitation of specific field
- what does this research say about the field and what has been published till now
- how do we keep the field reliable? which deviations should we try to avoid and which ones do not seem to matter much?

Conclusion

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